

TCGA research

This code performs survival analysis for the specified genes using TCGA data. Dr. Min Chen wants survival analysis for the following genes:

ITGB4

HIF1A

PTPRZ1

ITGB4 Analysis

Median Cutoff: From the plot of ITGB4 (median cutoff), we find a positive correlation with overall survival in the triple-negative subtype of breast cancer samples, with a p-value = 0.184. However, this correlation is not statistically significant.

25% and 75% Quantiles: From the plot of ITGB4 (25% and 75% quantiles), we find a positive correlation with overall survival, with a p-value = 0.579. This correlation is not statistically significant.

Best Cutoff: From the plot of ITGB4 (best cutoff), we find a positive correlation with overall survival, with a p-value = 0.67. This correlation is not statistically significant.

HIF1A Analysis

Median Cutoff: From the plot of HIF1A (median cutoff), we find a positive correlation with overall survival in the triple-negative subtype of breast cancer samples, with a p-value = 0.649. However, this correlation is not statistically significant.

25% and 75% Quantiles: From the plot of HIF1A (25% and 75% quantiles), we find a positive correlation with overall survival, with a

p-value = 0.461. This correlation is not statistically significant.

Best Cutoff: From the plot of HIF1A (best cutoff), we find a positive correlation with overall survival, with a p-value = 0.886. This correlation is not statistically significant.

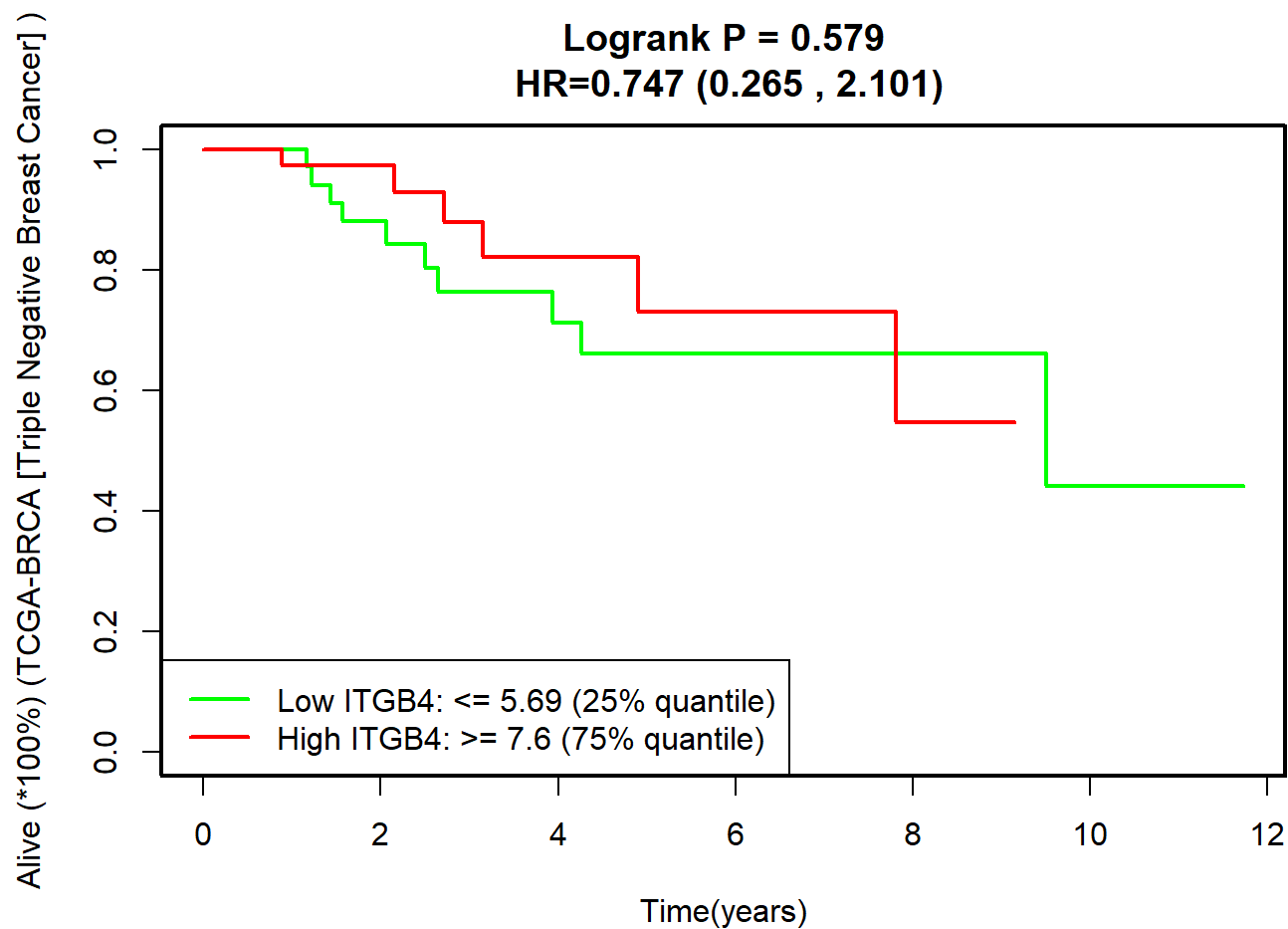
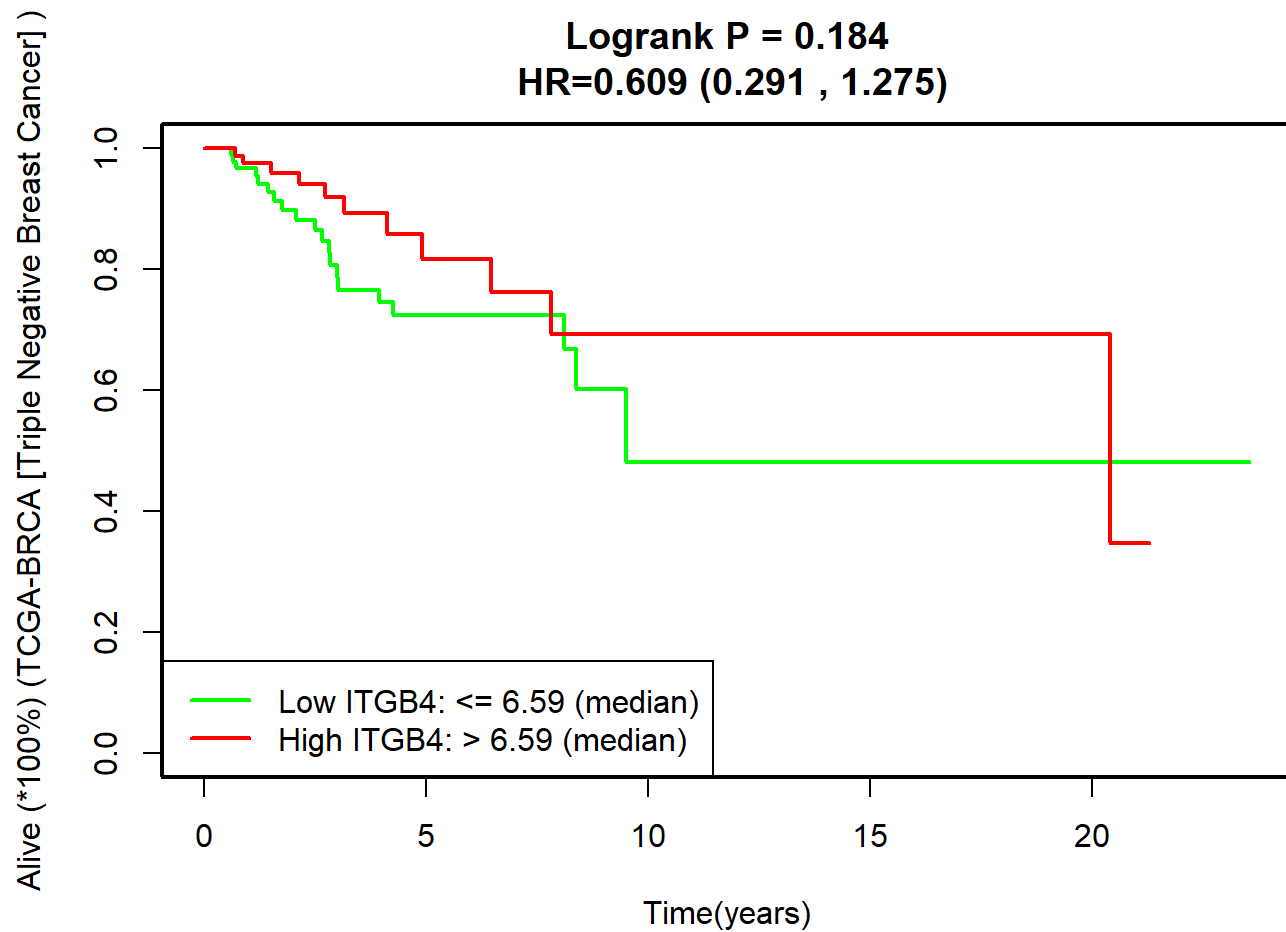
PTPRZ1 Analysis

Median Cutoff: From the plot of PTPRZ1 (median cutoff), we find a positive correlation with overall survival in the triple-negative subtype of breast cancer samples, with a p-value = 0.563. However, this correlation is not statistically significant.

25% and 75% Quantiles: From the plot of PTPRZ1 (25% and 75% quantiles), we find a positive correlation with overall survival, with a p-value = 0.839. This correlation is not statistically significant.

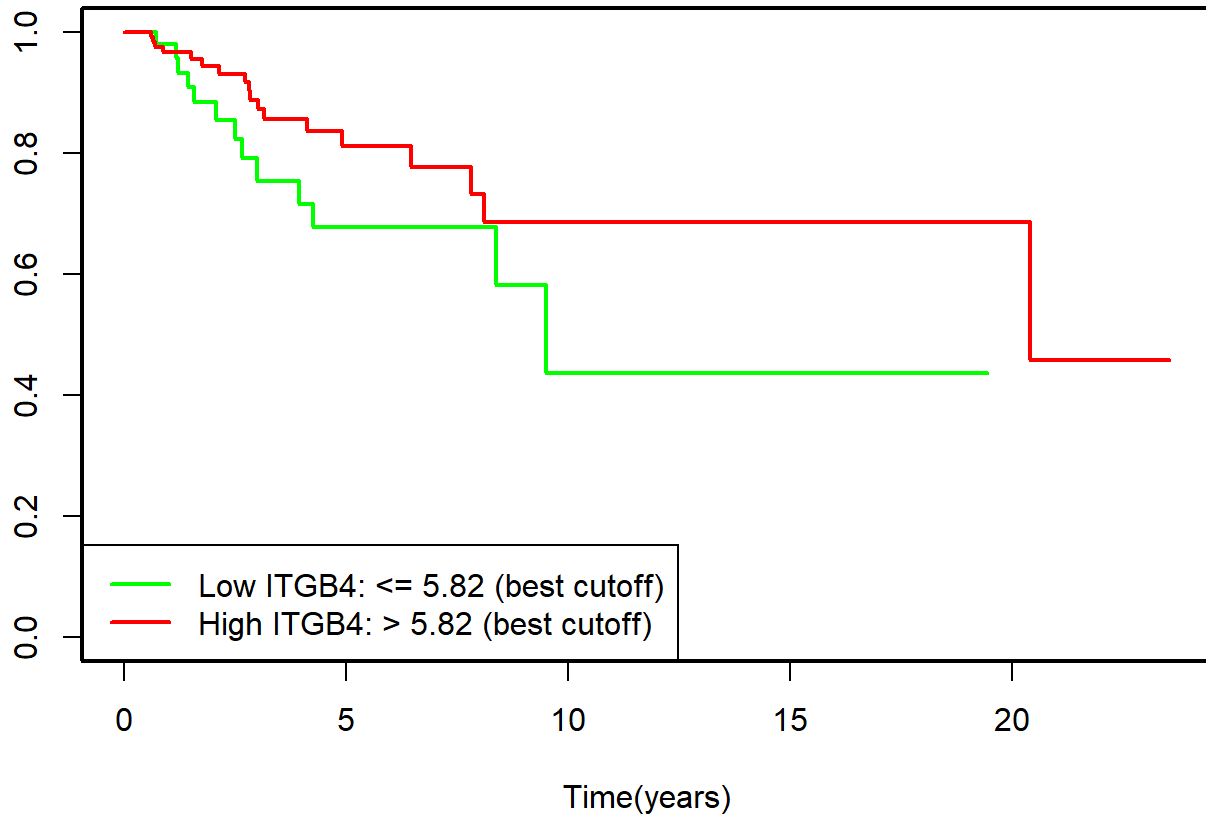
Best Cutoff: From the plot of PTPRZ1 (best cutoff), we find a positive correlation with overall survival, with a p-value = 0.772. This correlation is not statistically significant.

```
#>
#> Attaching package: 'dplyr'
#> The following objects are masked from 'package:stats':
#>
#>     filter, lag
#> The following objects are masked from 'package:base':
#>
#>     intersect, setdiff, setequal, union
```



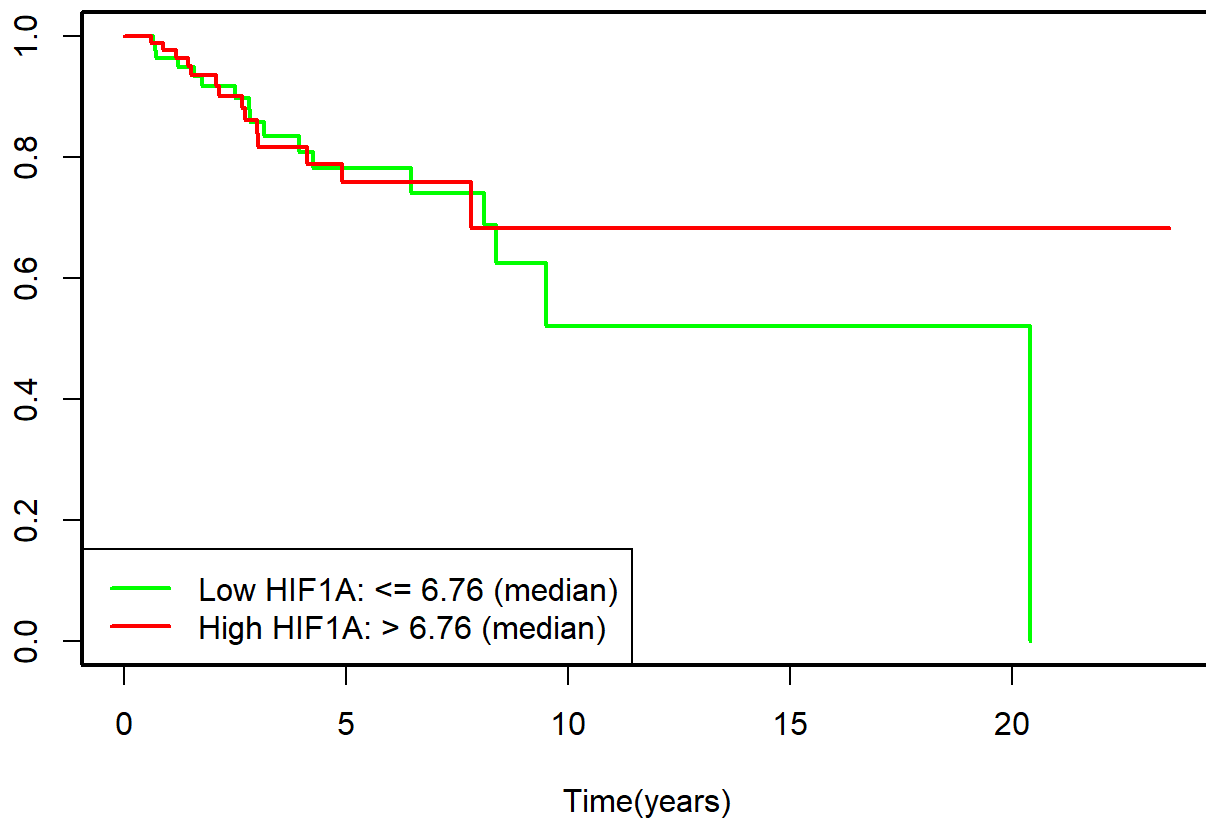
Alive (*100%) (TCGA-BRCA [Triple Negative Cancer])

Logrank P = 0.67
HR=0.554 (0.268 , 1.143)



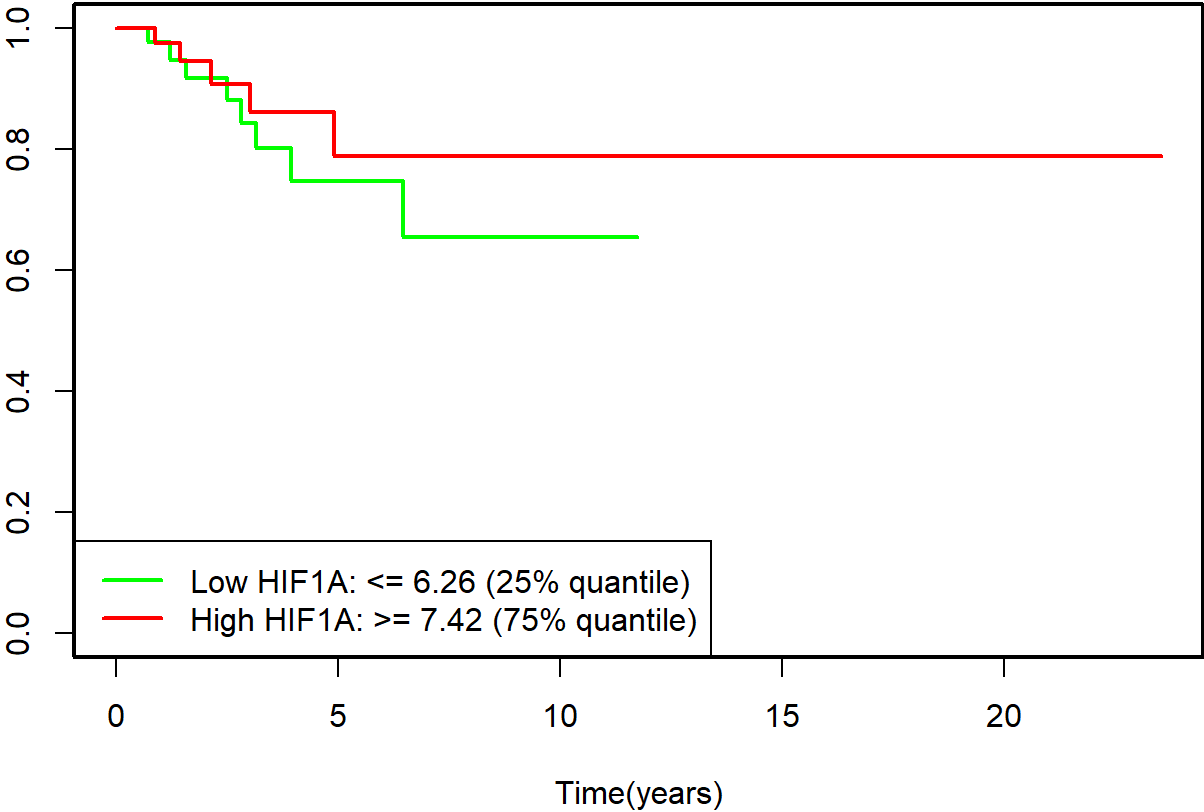
Alive (*100%) (TCGA-BRCA [Triple Negative Cancer])

Logrank P = 0.649
HR=0.848 (0.416 , 1.727)



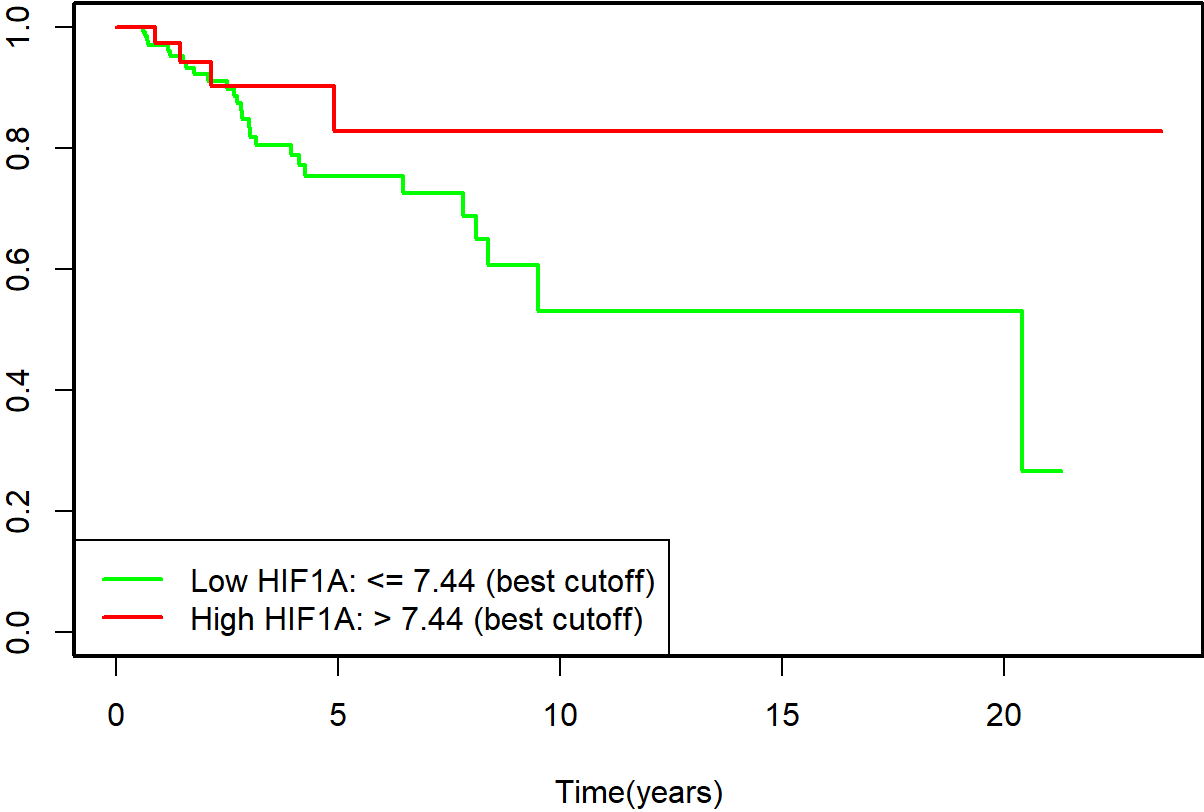
Alive (*100%) (TCGA-BRCA [Triple Negative Cancer])

Logrank P = 0.461
HR=0.659 (0.215 , 2.017)



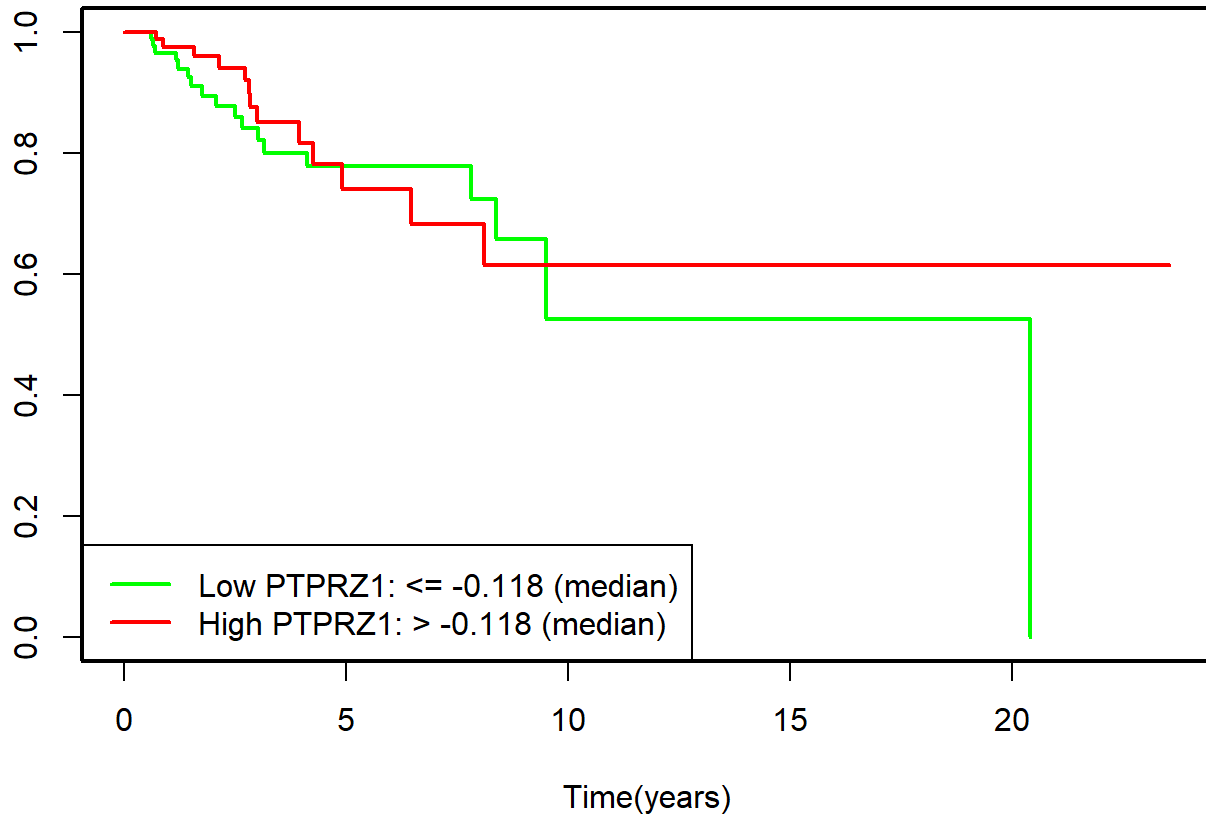
Alive (*100%) (TCGA-BRCA [Triple Negative Cancer])

Logrank P = 0.888
HR=0.509 (0.178 , 1.457)



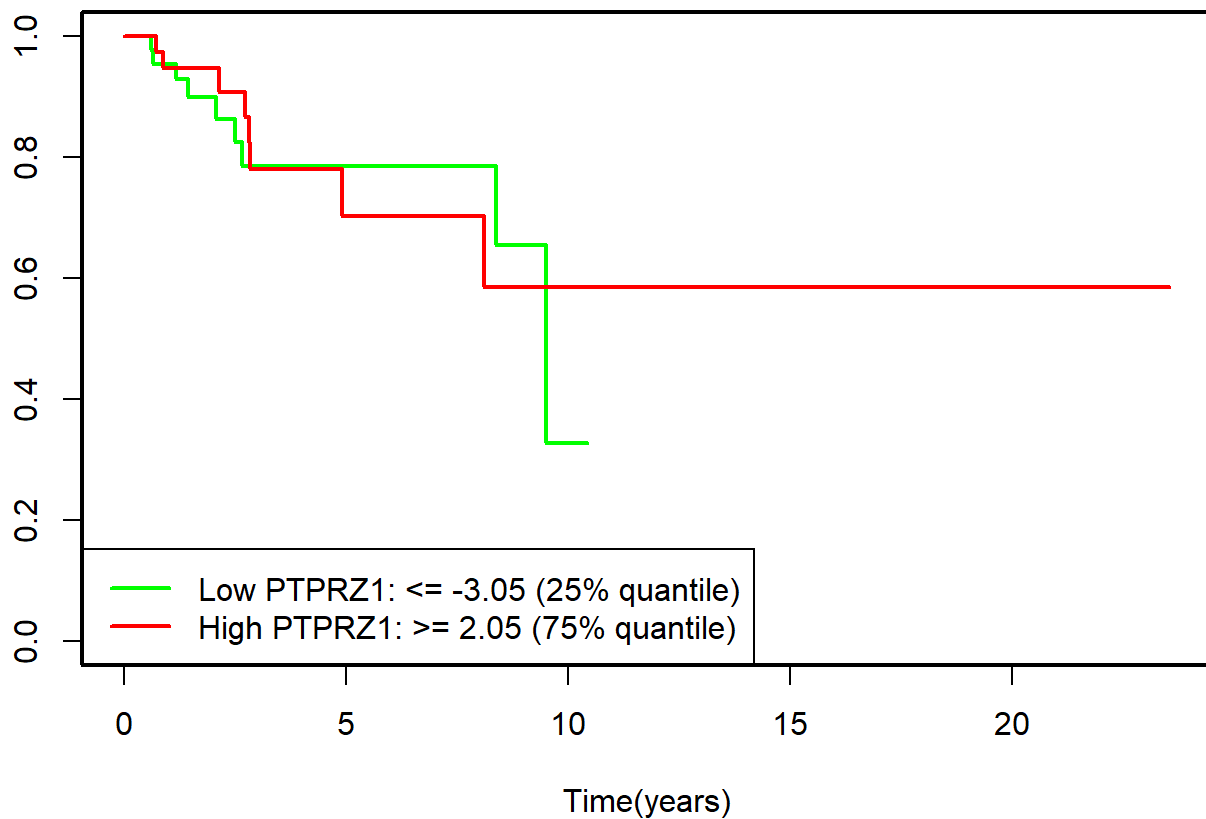
Alive (*100%) (TCGA-BRCA [Triple Negative Breast Cancer])

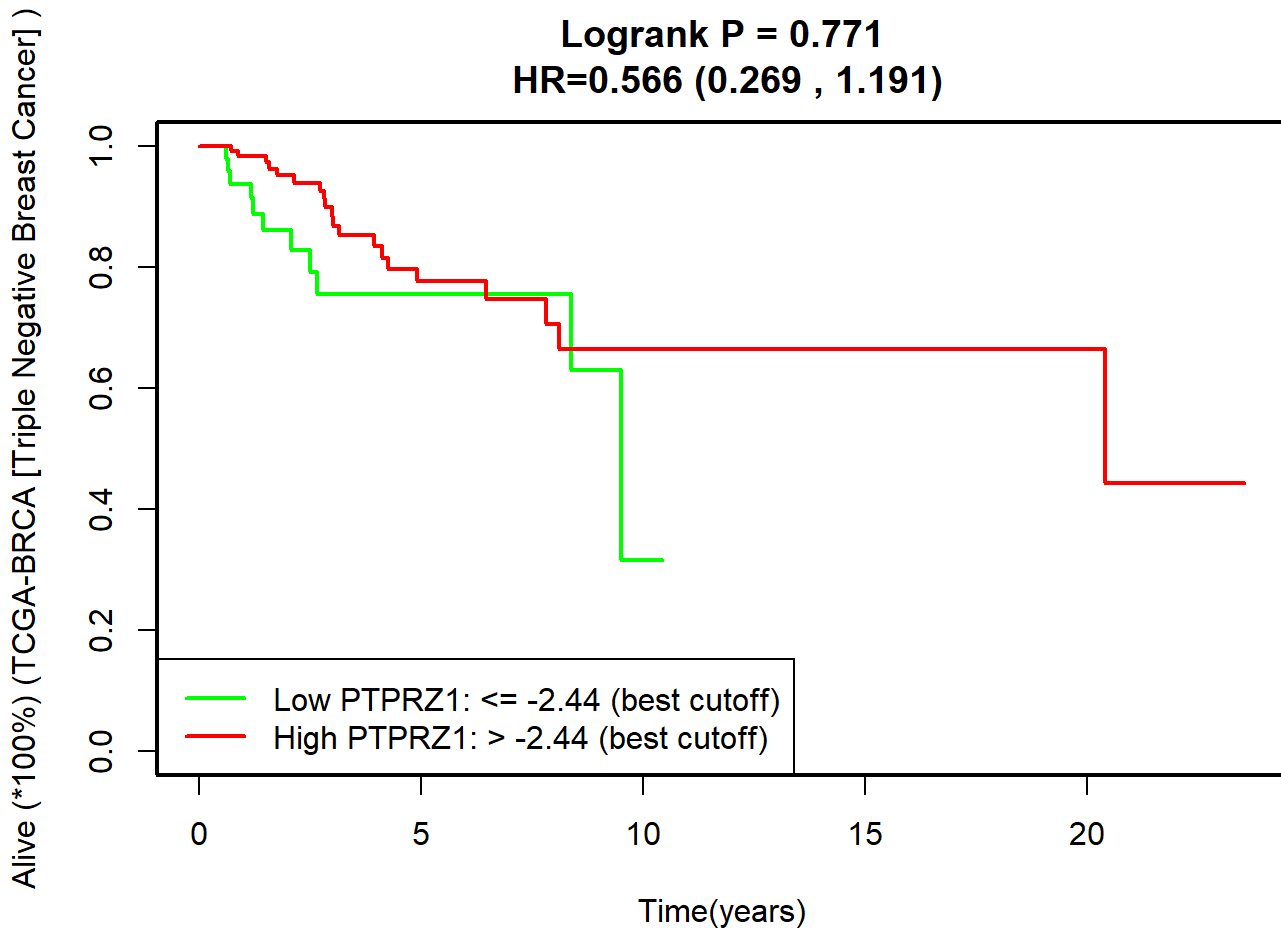
Logrank P = 0.563
HR=0.809 (0.395 , 1.659)



Alive (*100%) (TCGA-BRCA [Triple Negative Breast Cancer])

Logrank P = 0.839
HR=0.906 (0.349 , 2.351)





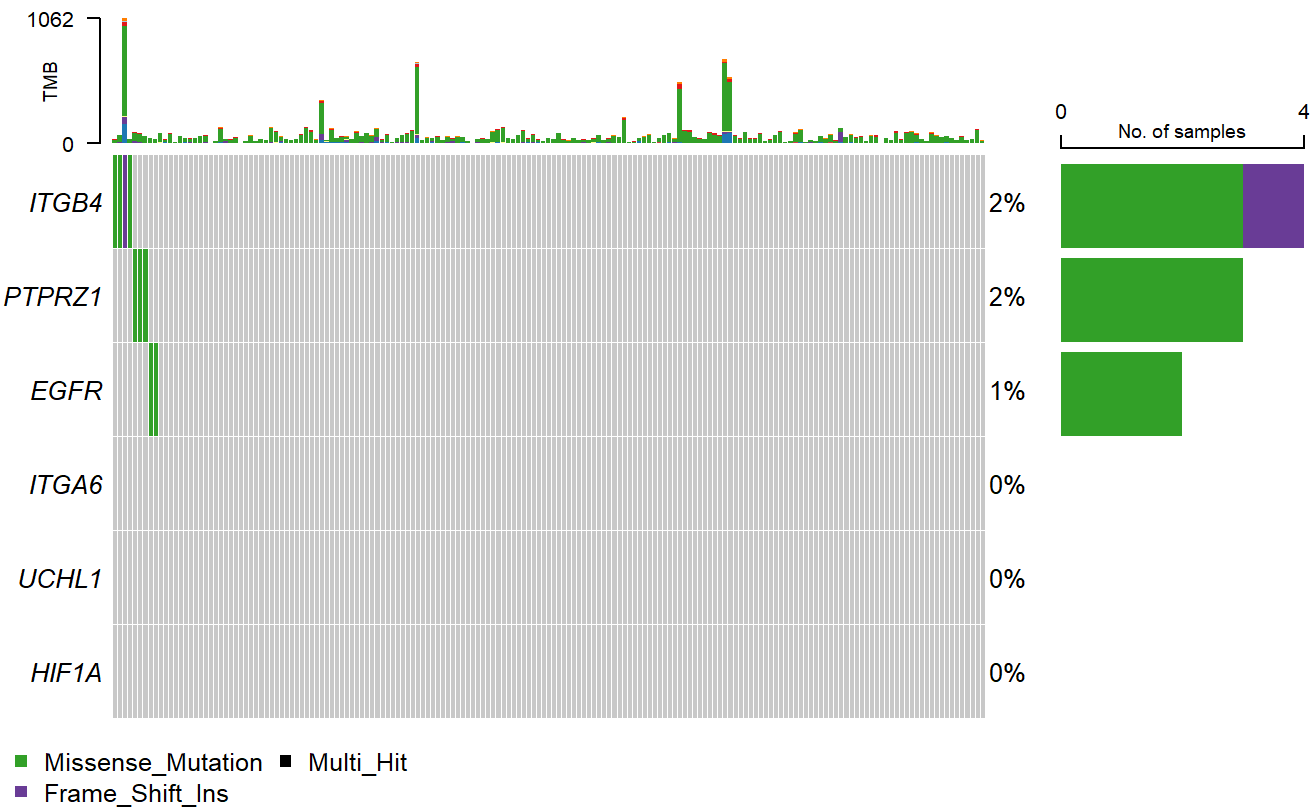
Dr.Min Chen is interested to see if there is co-occurrence of the mutations among the genes: ITGA6, ITGB4, UCHL1, H1F1A, PTPRZ1, EGFR within the TNBC subtype of breast cancer patients.

From the oncoplot, it seems that all the gene mutations specified by Dr.Chen are pretty rare, and we didn't find any strong evidence for co-occurrence

among these mutations.

```
#> -Reading
#> -Validating
#> --Removed 3485 duplicated variants
#> -Silent variants: 24868
#> -Summarizing
#> --Possible FLAGS among top ten genes:
#>   TTN
#>   MUC16
#> -Processing clinical data
#> --Missing clinical data
#> -Finished in 7.260s elapsed (2.390s cpu)
#> New names:
#> • `Sig_1_APOBEC` -> `Sig_1_APOBEC...11`
#> • `Sig_2_DEAMiN` -> `Sig_2_DEAMiN...12`
#> • `Sig_3_MMR` -> `Sig_3_MMR...13`
#> • `Sig_4_HR` -> `Sig_4_HR...14`
#> • `Sig_1_APOBEC` -> `Sig_1_APOBEC...125`
#> • `Sig_2_DEAMiN` -> `Sig_2_DEAMiN...126`
#> • `Sig_3_MMR` -> `Sig_3_MMR...127`
#> • `Sig_4_HR` -> `Sig_4_HR...128`
#> --Possible FLAGS among top ten genes:
#>   TTN
#>   USH2A
#>   MUC16
#>   SYNE1
#>   FLG
#> -Processing clinical data
```


Altered in 9 (5.2%) of 173 samples.



```
#>      gene1  gene2 pValue oddsRatio    00    01    11    10 pAdj
#>    <char> <char>  <num>      <num> <int> <int> <int> <int> <num>
#> 1:  ITGB4 PTPRZ1      1          0   166    3    0    4     1
#> 2:   EGFR ITGB4      1          0   167    4    0    2     1
#> 3:   EGFR PTPRZ1      1          0   168    3    0    2     1
#>
#>          Event      pair event_ratio
#>          <char>      <char>      <char>
#> 1: Mutually_Exclusive ITGB4, PTPRZ1      0/7
#> 2: Mutually_Exclusive  EGFR, ITGB4      0/6
#> 3: Mutually_Exclusive  EGFR, PTPRZ1      0/5
```